

2. Continue dividing. Bring down the 0 from the tens place in the dividend. How many times does 4 go into 20? Write the answer 5 directly above the second 0 in the dividend. Since $4 \times 5 = 20$, subtract $20 - 20 = 0$. Bring down the 6 from the ones place in the dividend. How many times does 4 go into 6? Write the answer 1 above the 6 in the dividend. Since $4 \times 1 = 4$, subtract $6 - 4 = 2$. Write the **remainder** of 2 as part of the quotient.

$$\begin{array}{r} 251 \text{ r}2 \\ 4 \overline{)1006} \\ \underline{8} \\ 20 \\ \underline{20} \\ 06 \\ \underline{04} \end{array}$$

By reviewing and memorizing multiplication tables, you can save yourself precious time on the GED® *Mathematical Reasoning Test*.

Practice 2.2

A. Solve.

1. $\begin{array}{r} 121 \\ \times 4 \\ \hline \end{array}$

4. $5 \overline{)65}$

7. $\begin{array}{r} 45 \\ \times 30 \\ \hline \end{array}$

10. $10 \overline{)280}$

2. $\begin{array}{r} 250 \\ \times 4 \\ \hline \end{array}$

5. $7 \overline{)735}$

8. $\begin{array}{r} 105 \\ \times 25 \\ \hline \end{array}$

11. $15 \overline{)225}$

3. $\begin{array}{r} 342 \\ \times 8 \\ \hline \end{array}$

6. $9 \overline{)189}$

9. $\begin{array}{r} 211 \\ \times 16 \\ \hline \end{array}$

12. $19 \overline{)114}$

B. Solve. If multiplying more than two numbers, find the product of two numbers before multiplying by the next number, and so on.

13. $50 \times 5 =$

16. $1004 \div 5 =$

19. $10,760 \div 20 =$

14. $179 \div 4 =$

17. $25 \times 3 \times 2 =$

20. $12 \times 8 \times 4 =$

15. $5 \times 6 \times 10 =$

18. $7452 \times 9 =$

21. $144,140 \div 12 =$

C. Choose the one best answer to each question.

22. A fruit juice container holds 16 servings. If the serving size is 6 ounces, how many ounces does the container hold in all?

- A. 10
- B. 22
- C. 76
- D. 96

D. For question 24, write your answer on the line provided.

24. How many 2-foot lengths can be cut from the string shown below?

12 ft

Answers and explanations begin on page 661.

23. A cashier has fifteen \$5 bills. How much money does he have?

- A. \$15
- B. \$25
- C. \$75
- D. \$150

Lesson 3: GED® Test Calculator Skills

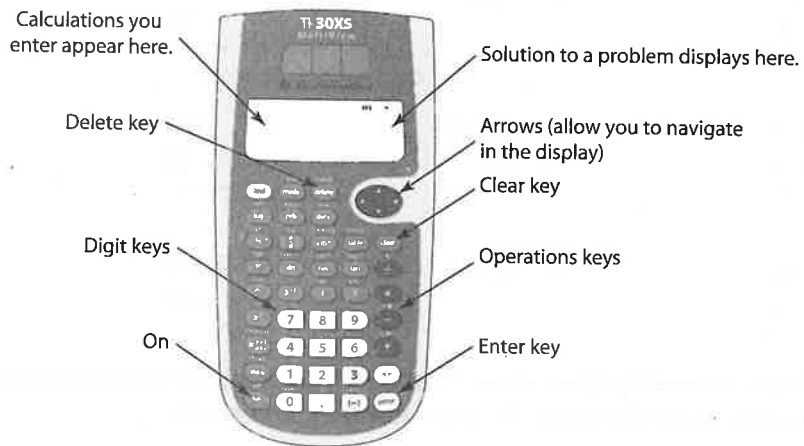
Key Ideas

- For many questions, you may use either a handheld or online version of the TI-30XS MultiView™ scientific calculator.
- Use the calculator when you think it will save you time or improve your accuracy.
- Learn the placement of the keys and how to perform specific functions before you take the test.

You will have access to an online or handheld calculator for many questions on the *Mathematical Reasoning Test* and for some questions on the *Social Studies and Science Tests*. If you wish to use a handheld calculator, you will likely need to bring your own. It must be the Texas Instruments TI-30XS MultiView™, shown here. The online version will be available when the calculator icon appears at the top of the computer screen. You will click on the icon to open the calculator and use the mouse to move it so that it does not cover the problem. See page 216 for more calculator basics.



Look at the reproduction of the calculator below and follow the examples that demonstrate how to use the calculator's operation keys.



GED® TEST TIP

You can buy a handheld TI-30XS MultiView™ calculator for practice as you work through this study guide. It may be available online, at office supply stores, or other places where calculators are sold.

Example 1: Add $63 + 97 + 58 + 32 + 81$.

1. Always clear a calculator before starting a new computation. On the TI-30XS MultiView™, use the *clear* key.

clear

2. Enter each number followed by the plus sign. As you type, the numbers and plus signs will appear on the calculator's screen. If you make a mistake, press *delete* to go back and reenter a number.

$63 + 97 + 58 + 32 + 81$

3. Press *enter* to find the total.

enter

The total, **331**, will appear on the right-hand side of the display:

331

Example 2: Find the difference between 15,789 and 9,332.

1. Always clear a calculator before starting a new computation.

clear

2. Enter the greater number first, followed by the minus operator.

15789 **=**

NOTE: Use the minus key that is on the right side of the calculator with the other operation symbols. Don't use the $(-)$ key at the bottom of the calculator; that key is used to enter a negative number.

3. Enter the number being subtracted.

9332

4. Press the *enter* key to find the answer.

enter

The answer **6457** will appear on the right-hand side of the display.

6457

Example 3: Find the product of 309 and 68.

1. Always clear a calculator before starting a new computation.

clear

2. Enter the first number, followed by the multiplication operator.

309 **×**

(The multiplication sign will appear in the display as an asterisk rather than an $\times$, but the *multiplication key* looks like an $\times$.)

3. Enter the next number.

68

4. Press the *enter* key to find the product.

enter

The answer **21012** will appear on the right-hand side of the display.

21012

Example 4: Divide 12,456 by 12.

1. Always clear a calculator before starting a new computation.

clear

2. Enter the number to be divided first, followed by the division operator.

12456 **÷**

3. Enter the number you are dividing by.

12

4. Press the *enter* key to find the quotient.

enter

The answer **1038** will appear on the right-hand side of the display.

1038

Practice 3.1

A. Practice solving the following problems on your calculator.

1. $19 + 26 + 85 + 23 =$

2. $2579 - 1392 =$

3. $4 \times 28 \times 7 =$

4. $2568 \div 107 =$

5. $12,356 + 14,728 =$

6. $107,899 - 93,457 =$

7. $209 \times 56 =$

8. $972 \div 18 =$

9. $20,540 \div 13 =$

B. Choose the one correct answer to each question.

10. Dan bought a used car with 16,741 miles on it. If the car now has 42,920 miles on it, how many miles has Dan put on the car?

A. 16,741

B. 26,179

C. 42,920

D. 59,661

11. A shipment of 20 computers arrived at a warehouse. If each computer is valued at \$995, what is the total value of the shipment?

A. \$995

B. \$1015

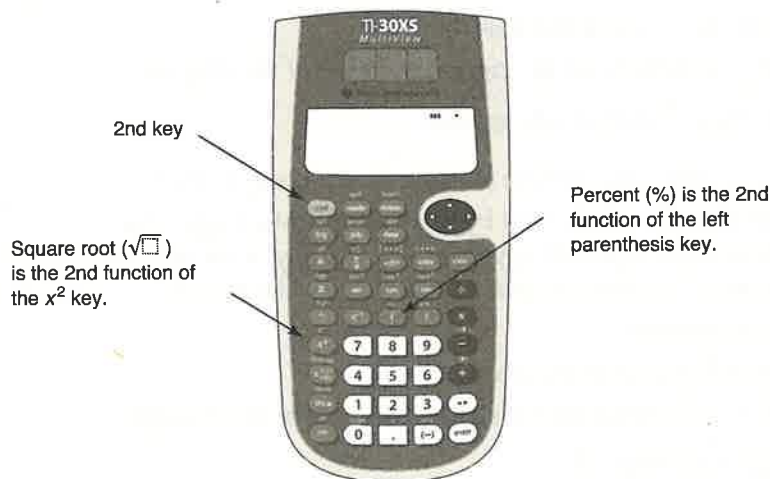
C. \$1990

D. \$19,900

Answers and explanations begin on page 661.

Using the 2nd Key for a Second Function

To access some of the functions on the TI-30XS MultiView™, you need to press the *2nd* key in the upper left corner of the keypad. This bright green key will activate the second function also shown in green above the corresponding key. To access the second function of a key, press the *2nd* key first—do not press it at the same time as the function key. Highlighted below are two commonly used second functions—square root and percent. Use the process shown here for all *2nd function* keys.



Example 5: Find the square root of 169.

1. Always clear a calculator before starting a new computation.
2. Recognize that square root is a *2nd* function over the x^2 key. Press the *2nd* key. (Note that the term *2nd* now appears in the upper left corner of the display.)
3. Next, press the x^2 key to activate the square root function in green over the key. You will see a blinking cursor under the square root.
4. Now that you have the square root function, enter the number. The number will appear under the square root symbol in the display.
5. Press the *enter* key to find the square root. The answer, 13, will appear on the right-hand side of the display.

Keys to Press

On the Display

clear

2nd

$\sqrt{\square}$

169

enter

2nd

$\sqrt{\square}$

$\sqrt{169}$

$\sqrt{169}$ 13

For more information about square roots, see page 320.

Remember: For all *2nd functions*, (1) press the *2nd* key first to activate the *2nd* function, (2) press the key, and (3) enter the numbers.

Example 6: Find the part if you are given the percent and the whole. Find 10 percent of 500.

1. Always clear a calculator before starting a new computation.
2. Enter the number you want to find the percent of.
3. Press the multiplication sign.
4. Enter the percent number.
5. Press the *2nd* key and then press the open, or left, parenthesis key to activate the percent function.
6. Press the *enter* key to find the answer.

clear
500
 $\times$
10
2nd %
enter
50

The answer, 50, will appear on the right-hand side of the display. Ten percent of 500 is 50.

Example 7: Find the percent if you are given the whole and the part. What percent of 240 is 60?

1. Always clear a calculator before starting a new computation.
2. Enter the part.
3. Press the division sign.
4. Enter the whole.
5. Press the *2nd* key and then press the close, or right, parenthesis symbol. This tells the calculator to translate the answer into a percent.
6. Press the *enter* key to find the answer.

clear
60
 $\div$
240
2nd %
enter
25%

The answer, 25%, will appear on the right-hand side of the display. Sixty is 25% of 240.

For more information about percents, see pages 264–275.

Practice 3.2

A. Practice solving the following problems on your calculator.

1. $\sqrt{625}$
2. $\sqrt{324}$
3. $\sqrt{1225}$
4. Find 20% of 680.
5. Find 10% of 1250.
6. 15 is what percent of 300?
7. Find 5% of 40.
8. Find 30% of 450.
9. 20 is what percent of 400?

B. Choose the one best answer to each question.

10. Tanya paid 20 percent of \$1680 as a down payment. How much was the down payment?
A. \$20
B. \$168
C. \$336
D. \$840
11. Aaron received a credit of \$48 on a purchase of \$960. What percent of \$960 is \$48?
A. 5%
B. 20%
C. 48%
D. 912%

Answers and explanations begin on page 661.